



# Introduction to Finance

## Capital Budgeting (Part I)

Professor: *Víctor Macías E.*

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# 1 What is an Investment Project?

An *investment project* is a structured action plan designed to solve a problem or capitalize on a market opportunity, ultimately satisfying a human or social need.

Its defining characteristic is the allocation of scarce resources (capital) in the present ( $I_0$ ) with the expectation of generating a stream of future net benefits ( $F_t$ ). To be viable, these benefits must exceed the initial investment after being discounted at a rate that reflects both the opportunity cost of capital and the associated risk.

## Key Examples of Investment Projects:

*Asset Replacement:* Replacing a gasoline vehicle with an electric one to reduce long-term operational costs.

*Market Expansion:* Launching a financial app that calculates loan payments to capture a new digital user base.

*Cost Optimization:* Installing solar panels to decrease dependency on the grid and lower energy expenses.

*Environmental Compliance:* Developing a plastic recycling plant to meet regulatory standards and sell repurposed materials.

*Process Improvement:* Implementing an automated inventory system for a retail business to minimize waste and stockouts.

*Human Capital Development:* Pursuing a Master in Finance to increase professional expertise and future earning potential.

# 2 Stages of an Investment Project

Three distinct stages are identified: pre-investment, investment, and operation.

## 2.1 Stage 1: Pre-investment

This is the formulation and evaluation phase of the project. Its objective is to determine the technical, economic, social, and environmental viability of the idea before committing significant capital.

The pre-investment stage involves the sequential and iterative development of the following levels of study. These aim to gradually reduce uncertainty regarding key variables (costs, benefits, and investment) to allow for a more rigorous financial evaluation at each step.

**i. Idea.** This includes a preliminary description of the project based on expert judgement or historical experience. It identifies customers, potential location, main investment items, a rough estimate of benefits and costs, and the project operator.

*Recommendation: Reject, postpone, or further develop the project idea by moving to the profile level.*

**ii. Profile.** A preliminary design of commercial, operational, administrative, and financial aspects is prepared using broad estimates of benefits and costs based on existing (secondary) information. An initial quantitative evaluation is performed using indicators such as the payback period and/or a preliminary NPV (Net Present Value) calculation, using a benchmark discount rate.

*Recommendation: Reject, postpone, or further develop the project by moving to a pre-feasibility study.*

**iii. Pre-feasibility.** This stage uses a combination of primary information (collected in the field) and secondary information. Market studies estimate revenue through a detailed analysis of industry supply and demand. Technical studies analyze and propose the project's size, technology, and location. Cash flows are constructed, and NPV and IRR (Internal Rate of Return) are calculated to evaluate solution alternatives. A sensitivity analysis is included to account for changes in the parameters with the greatest impact on profitability.

*Recommendation: Reject, postpone, or further develop the project by moving to a feasibility study, selecting and prioritizing viable alternatives based on the highest NPV.*

**iv. Feasibility.** This focuses on the precise measurement of costs and benefits for the single alternative deemed most viable in the pre-feasibility study. Investment and operating costs are based on formal quotes and price commitments. It includes a detailed investment schedule, equipment availability, and deadlines for delivery, installation, and testing. The human resources selection and training plan is finalized, and a detailed financing analysis (sources, conditions, and terms) is completed. This financial cash flow serves as the basis for the investment decision and the primary document for securing capital.

*Recommendation: Execution decision (investment) or definitive rejection of the project.*

## 2.2 Stage 2: Investment

Once the project is declared optimal and viable and the decision to invest has been made, this stage focuses on transforming the plan into a physical and operational reality. This is the period of the greatest capital expenditure ( $I_0$ ), where the bulk of the initial investment is incurred.

The activities include:

- i. Financial closing and legal management
  - Financing closing: Signing of credit agreements and the actual disbursement of committed funds, ensuring the availability of capital ( $I_0$ ) for execution.
  - Final permit management: Obtaining detailed construction licenses and the operational permits necessary for the start of works and assembly.
- ii. Acquisition and physical development
  - Fixed asset acquisition: Final purchase of land, critical machinery, primary equipment, technology, and patents.
  - Environmental implementation: Execution of Environmental Mitigation Works and compensation measures committed to during the Feasibility phase (e.g., treatment plants, reforestation).
  - Construction and engineering: Building of the physical infrastructure, equipment assembly, installation of production lines, and development of the remaining detailed engineering.
- iii. Testing and Operational Readiness
  - Commissioning (start-up): Preliminary operational tests, equipment calibration, and adjustment of production processes.
  - Training and organization: Final hiring and training of operational and administrative personnel.

## 2.3 Stage 3: Operation

This is the longest stage of the cycle. It begins when the production unit starts generating goods or services and net income flows ( $F_t$ ). The horizon of this stage is defined by the project's economic or technical useful life.

The key aspects are:

- Revenue Generation: Sale of products or services in the market.
- Cost Management: Handling of operating costs (labor, raw materials, maintenance).
- Ex-Post Evaluation (Control): Continuous monitoring to compare actual results ( $F_{actual}$ ) with the financial projections ( $F_{projected}$ ) made during the Pre-investment stage. In the event that significant deviations are identified, corrective actions must be applied.

### 3 Investment decision rules

*The objective of project evaluation is to identify those investments that increase the value of the firm.*

#### 3.1 Net present value (NPV)

The NPV is a measure of the additional wealth that an investment project generates for a company or investor, after having recovered the initial investment and covered the opportunity cost of capital.

The opportunity cost of capital is the expected rate of return offered by other assets with the same risk level as the project being evaluated.

The NPV is calculated using the following formula:

$$NPV = \sum_{t=1}^n \frac{Y_t}{(1+i_t)^t} - \sum_{t=1}^n \frac{E_t}{(1+i_t)^t} - I_0$$

where  $Y_t$  represents the income flows,  $E_t$  is the expenditure flow,  $I_0$  is the initial investment and  $i_t$  is the discount rate reflecting the opportunity cost. If this rate does not vary over time, then  $i_t = i \forall t$ .

The NPV measures value creation in current monetary units (dollars, euros, etc.); it is not a rate of return.

**The NPV is the preferred criterion in project evaluation, as it is the only criterion that directly measures value creation and leads to correct decisions in all contexts.**

*Criterion:*

- Accept the project if  $NPV \geq 0$ . This means the project recovers the initial investment and generates a return higher than (if  $NPV > 0$ ) or equal to (if  $NPV = 0$ ) the rate  $i$ . The project creates value. If  $NPV > 0$ , then if the owner were offered a buyout for the project, they would accept it only if the amount offered is greater than the calculated NPV.

*Accepting a project with a positive NPV is equivalent to increasing the market value of the company and, therefore, the wealth of its shareholders.*

- Reject the project if  $NPV < 0$ . This means the project does not reach the minimum return required by the rate  $i$  and, therefore, destroys value or causes the investor to miss the opportunity to earn that rate  $i$  in a similar investment.

*Excel function:* VNA()

NPV is not only useful for deciding whether or not to undertake a project; it can also be used to evaluate, among others, the following decisions:

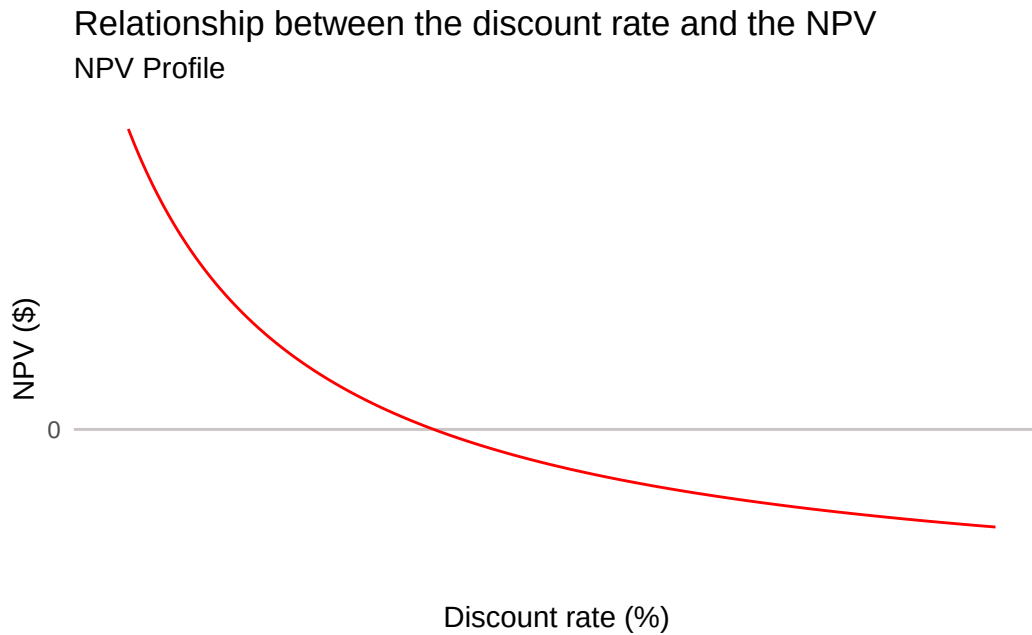
- What is the optimal location for a project?
- Which technology is the most convenient?
- Should a project be abandoned or continued?

**Example 1:**

Calculate the net present value of the cash flows  $F_1 = \$1800$  and  $F_2 = \$500$ , with an initial investment of 2000. Assume a discount rate of 10%

$$NPV = -2000 + \frac{1800}{(1 + 0.1)} + \frac{500}{(1 + 0.1)^2} = 50$$

Normally, in a project, the relationship between the discount rate and the NPV is inverse (See Example 6):



### Example 2:

Calculate the net present value of a cash flow of \$800 for a total of ten periods and an initial investment of 3000. Assume a discount rate of 10%.

$$NPV = -3000 + \sum_{t=1}^{10} \frac{800}{(1+0.1)^t} = -3000 + \frac{800}{0.1} \times \left[ 1 - \frac{1}{(1+0.1)^{10}} \right] = -3000 + 4915.65 = 1915.65$$

In Excel,

|    | A | B                      | C                              | D | E |
|----|---|------------------------|--------------------------------|---|---|
| 1  |   |                        |                                |   |   |
| 2  |   | Tasa de descuento (%)  | 10                             |   |   |
| 3  |   | Inversión inicial (\$) | 3000                           |   |   |
| 4  |   |                        |                                |   |   |
| 5  |   | Período                | Flujo                          |   |   |
| 6  |   | 0                      | -3000                          |   |   |
| 7  |   | 1                      | 800                            |   |   |
| 8  |   | 2                      | 800                            |   |   |
| 9  |   | 3                      | 800                            |   |   |
| 10 |   | 4                      | 800                            |   |   |
| 11 |   | 5                      | 800                            |   |   |
| 12 |   | 6                      | 800                            |   |   |
| 13 |   | 7                      | 800                            |   |   |
| 14 |   | 8                      | 800                            |   |   |
| 15 |   | 9                      | 800                            |   |   |
| 16 |   | 10                     | 800                            |   |   |
| 17 |   |                        |                                |   |   |
| 18 |   | VAN                    | =VNA(\$C\$2/100;C7:C16)+\$C\$6 |   |   |

### 3.2 Payback period

The number of years it takes to recover the initial investment. It can be calculated using either undiscounted (nominal) or discounted cash flows.

*Criterion:* Accept all projects that have a recovery period less than or equal to a specific threshold acceptable to the company. For example, accept a project if the investment is recovered in less than 3 years and reject it otherwise.

This criterion does not consider cash flows occurring after the payback period, subordinating acceptance to a liquidity factor rather than a profitability factor.

The payback period does not answer the question of how much value the project creates, but only how quickly the investment is recovered.

The *payback period (PP)* is defined as:

$$PP = \text{Year before full recovery} + \frac{\text{Unrecovered cost at start of year}}{\text{Discounted/Non-discounted cash flow during the year}}$$

**Example 3:**

The following table shows the cash flows (CF) of a project:

| Year | CF    |
|------|-------|
| 0    | -3000 |
| 1    | 500   |
| 2    | 700   |
| 3    | 800   |
| 4    | 1000  |
| 5    | 1200  |
| 6    | 1600  |

- (a) Calculate the payback period if the cash flows are not discounted.

The payback period is 4 years.

| Year | Non-discounted | Cumulative |
|------|----------------|------------|
| 0    | -3000          | -          |
| 1    | 500            | 500        |
| 2    | 700            | 1200       |
| 3    | 800            | 2000       |
| 4    | 1000           | 3000       |
| 5    | 1200           |            |
| 6    | 1600           |            |

- (b) Calculate the payback period if the cash flows are discounted at a rate of 10%.

As shown in the table, the cumulative PV turns positive during year 5. Specifically, at the end of year 4, there is still  $682.9 = 3000 - 2317.1$  to be recovered. In year 5, the project generates 745.1 in present value terms.

$$PP = 4 + \left( \frac{682.9}{745.1} \right) \approx 4.92 \text{ years}$$

The investment is recovered in approximately 4.92 years, which is close to 5 years.

| Year | Non-discounted | Discounted | Cumulative |
|------|----------------|------------|------------|
| 0    | -3000          | -          | -          |
| 1    | 500            | 454.5      | 454.5      |
| 2    | 700            | 578.5      | 1033       |
| 3    | 800            | 601.1      | 1634.1     |
| 4    | 1000           | 683        | 2317.1     |
| 5    | 1200           | 745.1      | 3062.5     |
| 6    | 1600           | 903.2      |            |

### 3.3 Internal rate of return (IRR)

It is the unique rate of return per period of a project, at which the total discounted benefits are exactly equal to the expenditures expressed in current currency.

It is the rate that makes  $NPV = 0$ . In other words, it is the threshold discount rate between the acceptance or rejection of a project. It represents the intrinsic average profitability of the project being evaluated. This profitability is not a market return, but rather an implicit rate derived from the project's cash flows. *Nothing guarantees that this rate is relevant for comparing projects or for reinvesting intermediate flows.*

$$\sum_{t=1}^n \frac{Y_t}{(1 + TIR)^t} - \sum_{t=1}^n \frac{E_t}{(1 + TIR)^t} - I_0 = 0$$

*Criterion:* Accept the project if  $IRR \geq i$ . Can you always apply this rule? (see *Example 6*) In other words, accept any project where the internal rate of return is greater than the discount rate.

*Question:* So far, it has been assumed that all flows are discounted at the same rate. However, if the curve representing the term structure of interest rates is not flat, then rates may differ during the evaluation period. In this case, with which discount rate should the IRR be compared?

*Excel function:* TIR()

#### Example 4:

Use the Excel *TIR* function to calculate the internal rate of return for the project included in *Example 3*. The IRR is 18.5%.

|    | A                      | B       | C            |
|----|------------------------|---------|--------------|
| 1  |                        |         |              |
| 2  | Inversión inicial (\$) |         | 3000         |
| 3  |                        |         |              |
| 4  |                        | Período | Flujo        |
| 5  |                        | 0       | -3000        |
| 6  |                        | 1       | 500          |
| 7  |                        | 2       | 700          |
| 8  |                        | 3       | 800          |
| 9  |                        | 4       | 1000         |
| 10 |                        | 5       | 1200         |
| 11 |                        | 6       | 1600         |
| 12 |                        |         |              |
| 13 | TIR                    |         | =TIR(C5:C11) |

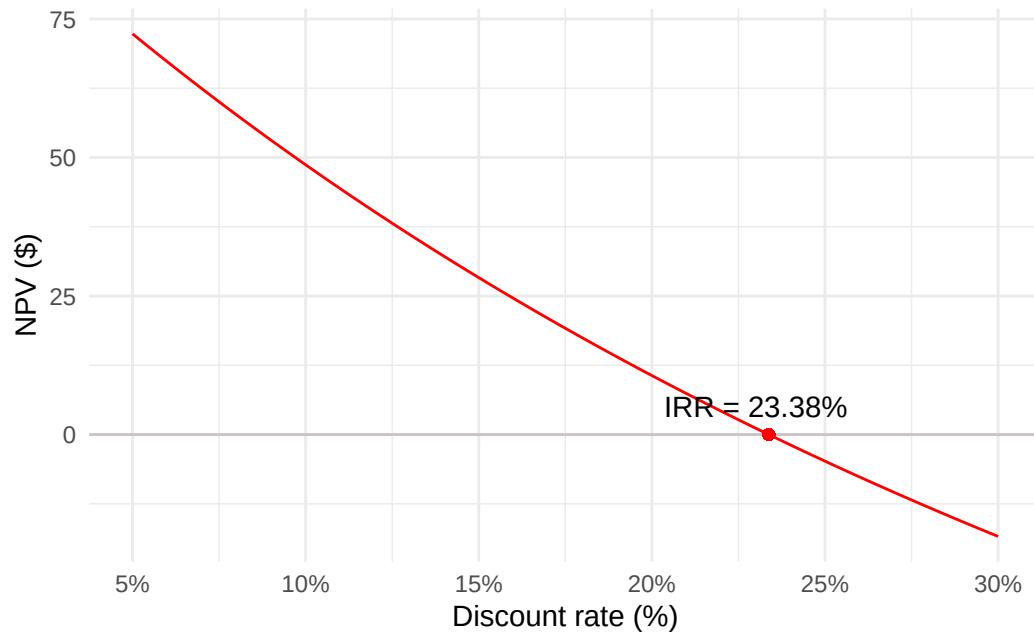
### Example 5:

Calculate the IRR of a project with an initial investment of \$200 and cash flows of \$100 for three periods.

$$-200 + \frac{100}{(1 + IRR)} + \frac{100}{(1 + IRR)^2} + \frac{100}{(1 + IRR)^3} = 0$$

Therefore, the  $IRR = 23.38\%$

Graphically,



Note that in this case:

- $IRR > r$  is equivalent to a  $NPV > 0$
- $IRR < r$  is equivalent to  $NPV < 0$
- $IRR = r$  is equivalent to a  $NPV = 0$

This relationship is observed only if the project is “normal,” meaning those projects where there is an initial capital outlay as an investment to obtain future positive flows.

**Example 6:**

Calculate the IRR of a project with an initial flow equal to \$200 and flows equal to  $-\$100$  for three periods.

$$200 - \frac{100}{(1 + TIR)} - \frac{100}{(1 + TIR)^2} - \frac{100}{(1 + TIR)^3} = 0$$

In this case, the  $IRR = 23.38\%$



Note that in this case:

- $IRR > r$  is equivalent to  $NPV < 0$
- $IRR < r$  is equivalent to  $NPV > 0$
- $IRR = r$  is equivalent to  $NPV = 0$

*Question: Use Generative AI to find examples of projects that have a positive initial flow and, subsequently, a combination of positive and negative flows.*

**Example 7:**

Calculate the IRR of a project with an initial investment of \$250 and perpetual cash flows of \$35.

$$NPV = -250 + \frac{35}{TIR}$$

Therefore, the IRR is 14%



### 3.4 Profitability index (PI)

The PI is calculated as the present value of future cash flows divided by the initial investment:

$$PI = \frac{\sum_{t=1}^n \frac{F_t}{(1+i)^t}}{I_0} = 1 + \frac{NPV}{I_0}$$

where  $NPV = -I_0 + \sum_{t=1}^n \frac{F_t}{(1+i)^t}$

#### Example 8:

Calculate the profitability index.

| Project | 0     | 1    | 2    | 3      |
|---------|-------|------|------|--------|
| A       | -\$20 | \$70 | \$10 | -\$120 |
| B       | -\$10 | \$15 | \$40 | -\$100 |

Assume a discount rate of 10%.

*Project A:*

$$\sum_{t=1}^n \frac{F_t}{(1+i)^t} = \frac{70}{1.12} + \frac{10}{1.12^2} = 70.47 \Rightarrow PI_A = \frac{70.47}{20} = 3.52$$

$$NPV_A = -20 + \frac{70}{1.12} + \frac{10}{1.12^2} = 50.47 \Rightarrow PI_A = 1 + \frac{50.47}{20} = 3.52$$

*Project B:*

$$\sum_{t=1}^n \frac{F_t}{(1+i)^t} = \frac{15}{1.12} + \frac{40}{1.12^2} = 45.28 \Rightarrow PI_B = \frac{45.28}{10} = 4.53$$

$$NPV_B = -10 + \frac{15}{1.12} + \frac{40}{1.12^2} = 35.28 \Rightarrow PI_B = 1 + \frac{35.28}{10} = 4.53$$

Which project to accept?

*Case 1: Independent projects.*

- Accept an independent project if  $PI > 1$  (This is equivalent to  $NPV > 0$ ).
- Reject if  $PI < 1$  (This is equivalent to  $NPV < 0$ ).

Therefore, the company must accept Projects A and B.

*Case 2: Mutually exclusive projects.* The fact that projects are mutually exclusive means that you can accept either one of them, or reject both, but they cannot be accepted simultaneously. In mutually exclusive projects, the problem is not evaluating projects individually, but choosing the alternative that maximizes value.

If the company can only accept one of its projects, we have:

Since  $NPV_A > NPV_B$ , the company should prioritize Project A, even though  $PI_A < PI_B$ . In this instance, the Profitability Index fails due to the scale problem, as it ignores the larger absolute value created by Project A's higher investment.

*Note: If we evaluate the incremental cash flows using the PI, the decision aligns with the NPV rule, favoring Project A (See Section 4.2 on incremental analysis)*

*Case 3: Capital rationing.* Capital rationing occurs when a firm's investment capital is insufficient to fund every project with a positive NPV. Under these constraints, ranking projects by absolute NPV can be misleading; instead, we rely on the Profitability Index (PI) to maximize value creation per dollar spent under certain conditions.

Consider a scenario with a \$20 budget limit and the following three projects:

| Project | 0     | 1    | 2    | 3      | 4      | NPV   | PI   |
|---------|-------|------|------|--------|--------|-------|------|
| A       | -\$20 | \$70 | \$10 | -\$120 |        | 50.47 | 3.52 |
| B       | -\$10 | \$15 | \$40 | -\$100 | -\$100 | 35.28 | 4.53 |
| C       | -\$10 | -\$5 | \$60 |        |        | 33.37 | 4.34 |

- With a \$20 budget, the firm cannot afford to fund both Project A and Project B.
- While Project A has the highest individual NPV (\$50.47), it consumes the entire budget.
- By selecting Projects B and C, the firm stays within the \$20 limit and generates a combined NPV of \$68.65 ( $35.28 + 33.37$ ). Although Project A is the superior standalone investment, the combination of Projects B and C—which both boast higher PIs—yields a higher total increase in firm value.

If the budget is \$30, does the decision change?

If the budget increases to \$30, the decision-making landscape shifts. The firm is no longer forced to choose between the high-efficiency smaller projects and the high-value larger ones; it can now afford the combination of Projects A and B.

Since  $NPV_A + NPV_B > NPV_B + NPV_C$ , the firm maximizes shareholder wealth by prioritizing the two highest-NPV projects. Relying solely on the Profitability Index (PI) ranking in this scenario would lead to a “suboptimal” decision, as it would overlook the absolute value created by the larger investment.

This logic assumes project indivisibility—the reality that most capital investments are “all or nothing.” Since we cannot implement 50% of a factory or a software system, we must evaluate projects as discrete units. Under these conditions, the goal is simply to find the specific “package” of projects that fits within the budget while producing the highest possible total NPV.

### 3.5 Equivalent annual benefit (EAB) and Equivalent annual cost (EAC)

The Equivalent Annual Benefit (EAB) represents the constant annual cash flow that provides the same present value as the project’s NPV. Under this method, the analyst compares projects with different lifespans by selecting the one that offers the highest annualized value.

When the analysis focuses exclusively on expenses rather than revenues, the approach is specifically termed the Equivalent Annual Cost method. In this case, EAC is a way to convert the total cost of a project into an annual cost, so you can compare projects with different lifespans.

To determine the Equivalent Annual Cost (EAC), follow this three-step process:

*Step 1: Determine the Present Value of Costs.*

Calculate the total PV of all outflows, including the initial investment ( $I_0$ ) and the present value of all operating and maintenance expenditures ( $E_t$ ) over the asset’s life ( $n$ ):

$$PV_{\text{Costs}} = I_0 + \sum_{t=1}^n \frac{E_t}{(1+i)^t}$$

*Step 2: Annualize the Total Cost.*

Convert the total present value into an annuity by dividing it by the annuity factor for the project's duration:

$$EAC = \frac{PV_{\text{Costs}}}{\frac{1}{r} \times \left[1 - \frac{1}{(1+r)^n}\right]}$$

*Step 3: Select the Optimal Project.*

Compare the results and choose the investment that yields the lowest EAC.

*Note: Follow the same steps to calculate the EAB (Equivalent Annual Benefit). Carefully evaluate the interpretation of results to ensure the selection of the most viable project.*

**Example 9:**

A utility company is choosing between two independent solar technologies for a new site. Both have a cost of capital of 10%.

*Technology A (Standard Panels):*

- Investment ( $I_0$ ): \$10 million
- Annual cash flow: \$2.5 million
- Life: 6 years

*Technology B (Premium High-Efficiency Panels):*

- Investment ( $I_0$ ): \$18 million
- Annual cash flow: \$4.0 million
- Life: 10 years

**Technology A**

$$NPV_A = -10 + \frac{2.5}{0.1} \times \left[1 - \frac{1}{(1 + 0.1)^6}\right] = 0.888 \text{ million}$$

$$EAB_A = \frac{NPV_A \times r}{\left[1 - \frac{1}{(1+r)^N}\right]} = \frac{0.888 \times 0.1}{\left[1 - \frac{1}{(1+0.1)^6}\right]} = 0.204 \text{ million per year}$$

## Technology B

$$NPV_B = -18 + \frac{4}{0.1} \times \left[ 1 - \frac{1}{(1 + 0.1)^{10}} \right] = 6.578 \text{ million}$$

$$EAB_B = \frac{NPV_B \times r}{\left[ 1 - \frac{1}{(1+r)^N} \right]} = \frac{6.578 \times 0.1}{\left[ 1 - \frac{1}{(1+0.1)^{10}} \right]} = 1.071 \text{ million per year}$$

The company should choose Technology B. While Technology B requires a larger upfront investment, it utilizes the site more effectively over a longer period, creating significantly more value for the utility company every year it is in operation.

### Example 10:

A company must choose between two machines. Revenues per year are assumed to be the same, regardless of machine, so they are not considered in the analysis. The aftertax cash flows from the two machines are shown here:

| Machine | 0      | 1      | 2      | 3      |
|---------|--------|--------|--------|--------|
| A       | -\$500 | -\$120 | -\$120 | -\$120 |
| B       | -\$600 | -\$100 | -\$100 | -\$100 |

Assume a discount rate of 10%.

### Machine A

$$PV_A = -500 - \frac{120}{1.1} - \frac{120}{1.1^2} - \frac{120}{1.1^3} = -500 - \frac{120}{0.1} \times \left[ 1 - \frac{1}{1.1^3} \right] = -798.42$$

The equivalent annual cost is:

$$EAC_A = \frac{PV_A \times r}{\left[ 1 - \frac{1}{(1+r)^N} \right]} = \frac{798.42 \times 0.1}{\left[ 1 - \frac{1}{(1+0.1)^3} \right]} = 321.06$$

### Machine B

$$PV_B = -600 - \frac{100}{1.1} - \frac{100}{1.1^2} - \frac{100}{1.1^3} - \frac{100}{1.1^4} = -600 - \frac{100}{0.1} \times \left[ 1 - \frac{1}{1.1^4} \right] = -916.99$$

The equivalent annual cost is:

$$EAC_B = \frac{PV_B \times r}{\left[1 - \frac{1}{(1+r)^N}\right]} = \frac{916.99 \times 0.1}{\left[1 - \frac{1}{(1+0.1)^4}\right]} = 289.28$$

Since  $EAC_B < EAC_A$  machine B is the preferred choice.

## 4 Issues with the IRR

### 4.1 Individual Projects

#### 4.1.1 Multiple IRRs

A project can have multiple Internal Rates of Return (IRR) when the project's cash flow changes sign more than once throughout its useful life. For example, the project may require a new investment or a high decommissioning cost in intermediate or final periods.

**Example 11:**

Calculate the IRR of a project with an initial investment of \$100 and cash flows equal to \$230 and  $-\$132$ .

In this specific case, because the signs change from negative to positive and back to negative, we are dealing with a quadratic equation. To find the IRR, we set the Net Present Value (NPV) to zero:

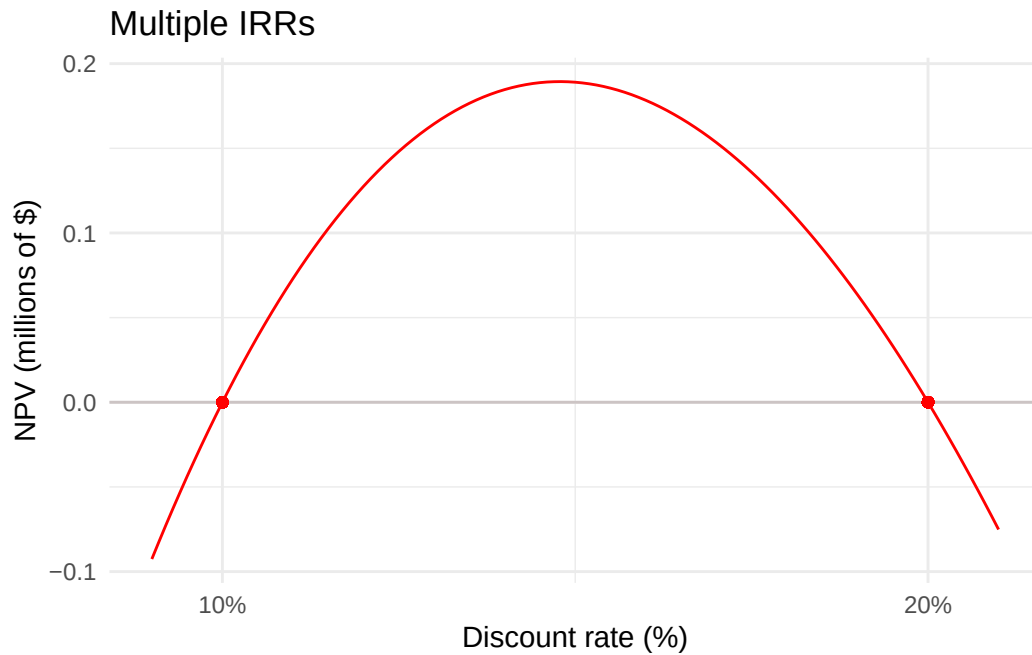
$$-100 + \frac{230}{(1 + IRR)} - \frac{132}{(1 + IRR)^2} = 0$$

If you solve this, you'll find that both 10% and 20% are technically correct IRRs for this project. This is a classic example of why the IRR can be a bit of a “wild card” when cash flows aren't conventional.

*Remember!* If the equation is  $ax^2 + bx + c = 0$ , then the solution is given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Graphically,



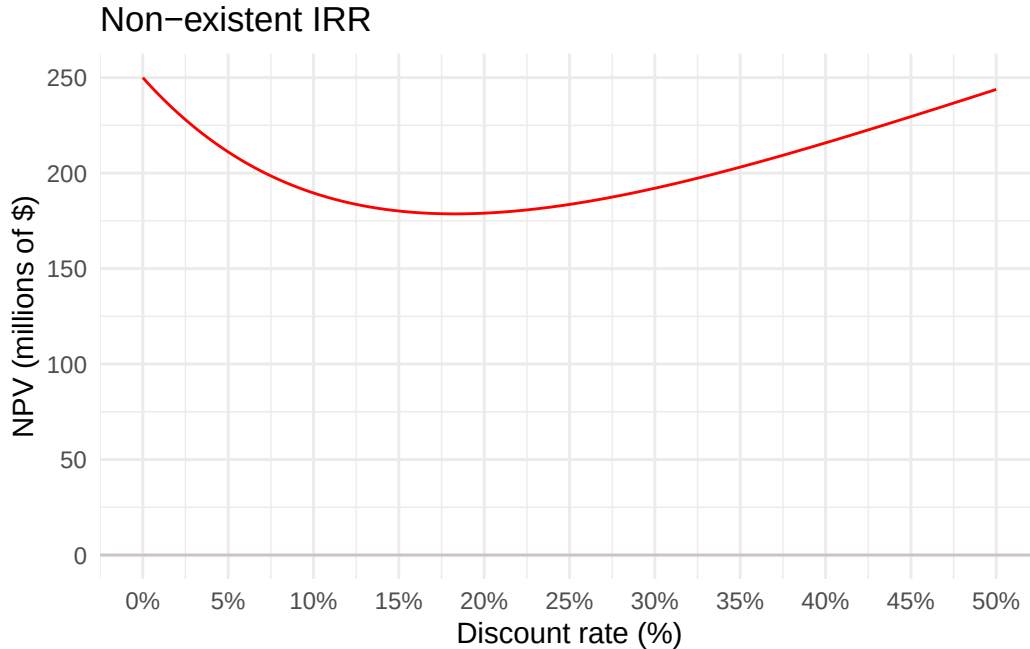
#### 4.1.2 Non-existent IRR

These are cases where the  $NPV$  is always positive or always negative.

##### Example 12:

Calculate the IRR of a project with an initial flow of \$750,000 and flows in the remaining periods equal to  $-\$500,000$ ,  $-\$500,000$ ,  $-\$500,000$ , and  $\$1,000,000$ .

$$NPV = 750000 + \frac{-500000}{(1+r)} + \frac{-500000}{(1+r)^2} + \frac{-500000}{(1+r)^3} + \frac{1000000}{(1+r)^4}$$



In this case, the NPV is always greater than zero.

- If the discount rate is zero, then the  $NPV = \$250.000$ .
- If the discount rate tends toward infinity, the NPV tends toward  $\$750.000$ .

No matter how much the discount rate is increased, the negative flows are small enough compared to the initial flow and the terminal value that the project never stops being profitable.

#### 4.1.3 Conditions for IRR to be a Valid Criterion

For the IRR to be a reliable metric in decision-making, the following requirements must be met:

##### 1. *Change of sign*

For at least one real IRR to exist, the project's cash flow must show at least one change of sign. If all flows are positive (or all negative), the NPV function will never cross the horizontal axis (where  $NPV = 0$ ).

Generally, this means having an initial investment (negative flow) followed by benefits (positive flows).

##### 2. *Conventional cash flow*

The IRR will be unique if and only if the project has a conventional cash flow.

*A flow is conventional when there is exactly one sign change in the entire time series. Typically:  $(-, +, +, +, \dots)$*

If the project has “non-conventional” flows—meaning there is more than one sign change—multiple IRRs could exist.

In theory, a cash flow with  $k$  sign changes can have up to  $k$  internal rates of return. In these cases, NPV should be used as the acceptance criterion.

### *3. Reinvestment assumption*

The IRR rule assumes that intermediate cash flows can be reinvested at the IRR itself.

## **4.2 Comparison between Mutually Exclusive Projects**

If two projects are mutually exclusive, the NPV and IRR criteria may provide opposing recommendations. However, NPV should be preferred.

The correct decision is based on the NPV of the incremental cash flows.

*Incremental cash flows measure the opportunity cost of choosing one project instead of its best alternative. Choosing between mutually exclusive projects is deciding whether it is worth paying the incremental cost of moving from one alternative to another. The base project represents the opportunity cost of the decision.*

*If the NPV of the incremental flow  $(A - B)$  is positive, then A dominates B and should be chosen. In mutually exclusive projects, the incremental flow does not measure if a project is good in the abstract, but whether it is better than its best alternative.*

### **4.2.1 Projects that Differ in Scale**

When projects have different sizes, the IRR can be misleading. A small project might have a massive percentage return (IRR), but a much larger project with a lower IRR might actually generate more total wealth (NPV).

In these cases, we often look at Incremental IRR—the internal rate of return on the additional money spent to choose the larger project over the smaller one.

#### **Example 13:**

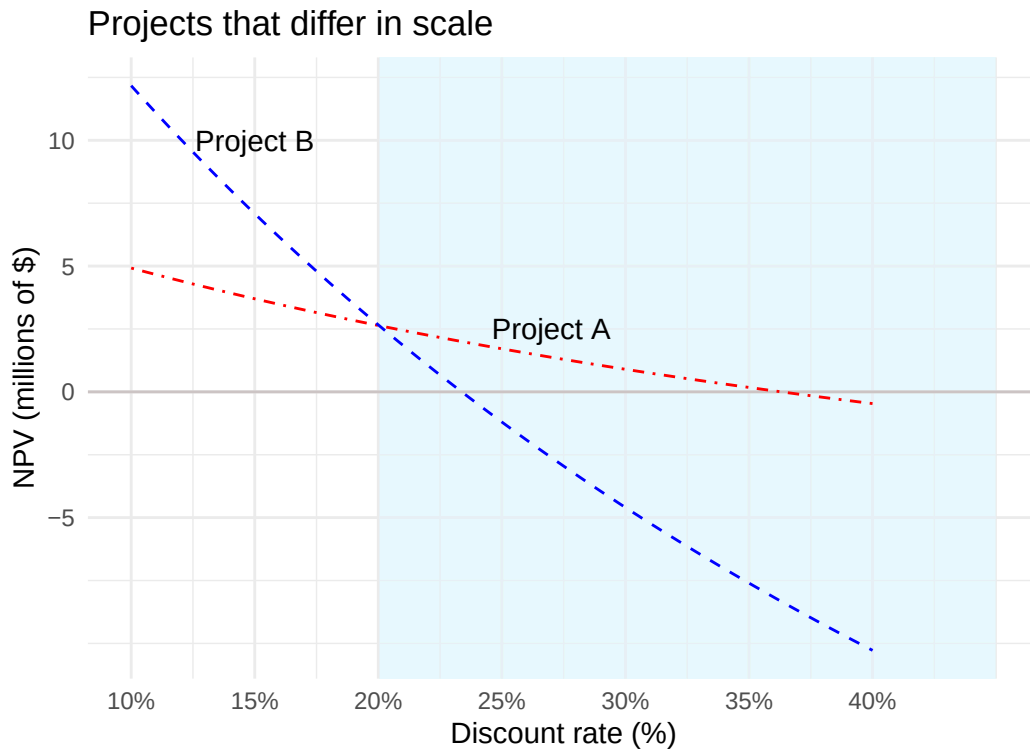
The following are the cash flows for two projects that differ significantly in both their initial investment and their subsequent flows:

| Project | 0   | 1  | 2  | 3  |
|---------|-----|----|----|----|
| A       | -10 | 6  | 6  | 6  |
| B       | -50 | 25 | 25 | 25 |

- (a) Calculate the IRR and the NPV for both projects, assuming a discount rate of 12% and 22%. What can you conclude?

|                    | Project A | Project B |
|--------------------|-----------|-----------|
| IRR                | 36.3%     | 23.4%     |
| NPV ( $r = 12\%$ ) | 4.41      | 10.05     |
| NPV ( $r = 22\%$ ) | 2.25      | 1.06      |

- (b) Graph the relationship between the NPV and discount rates for projects A and B.



As shown in the graph:

- If the discount rate is 12%, then Project B is chosen because it has a higher NPV, even though it has a lower IRR.

- If the discount rate is greater than 20%, Project A has a higher NPV.
- (c) Calculate the IRR and NPV (assuming discount rate is 12%) of the incremental cash flows.

| Project   | 0   | 1  | 2  | 3  |
|-----------|-----|----|----|----|
| A         | -10 | 6  | 6  | 6  |
| B         | -50 | 25 | 25 | 25 |
| Increment | -40 | 19 | 19 | 19 |

Note that the incremental flows are calculated using Project A as the base, which is the project with the lower initial investment.

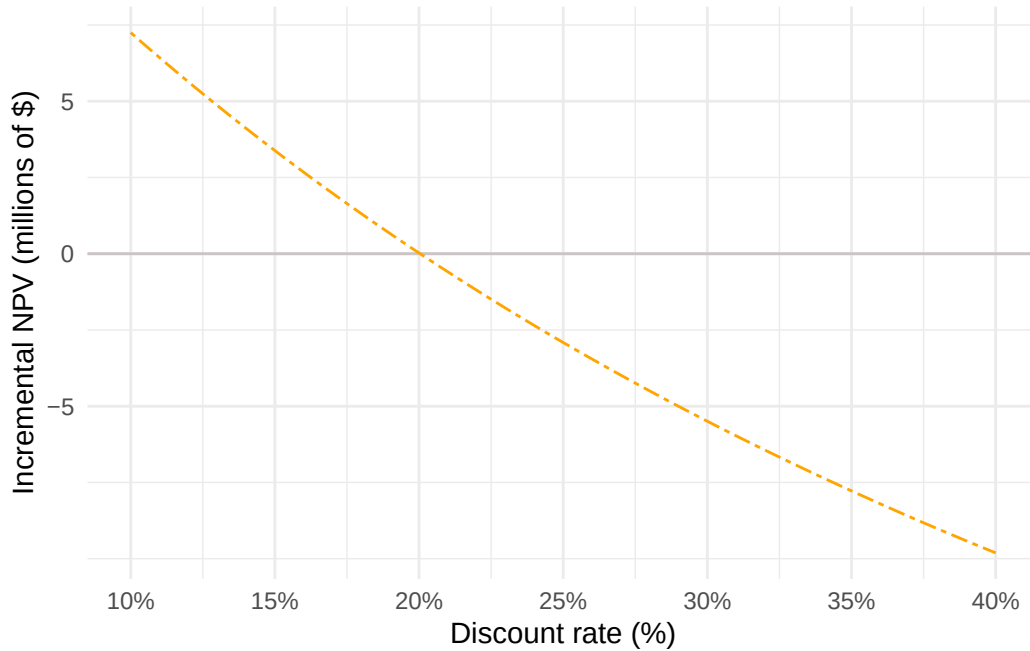
The IRR of the incremental flows is 20.04%.

The NPV of the incremental cash flows is:

$$NPV = -40 + \frac{19}{1.12} + \frac{19}{1.12^2} + \frac{19}{1.12^3} = 5.63$$

- The calculated incremental IRR is 20.04%.
- The NPV of the incremental cash flows is \$5.63
- At any discount rate below 20.04%, Project B (the larger scale project) will always have a higher NPV.
- As soon as the discount rate exceeds 20.04%, the advantage shifts to Project A (the smaller scale project) because its value is less sensitive to the cost of capital.

**The incremental IRR identifies the point at which the decision changes, but it does not measure value creation. For this reason, the final decision must always be based on the incremental NPV.**



*The incremental NPV must be discounted at the rate that reflects the risk of the incremental cash flows. If the projects have similar risks, this rate coincides with the common cost of capital; if the risks differ, it must be adjusted to reflect the risk of the additional business.*

*The correct rate does not depend on the project chosen, but rather on the risk of the cash flows that are changing.*

(d) If the relevant discount rate is 16%, what decision should be made?

If the discount rate is 16%, then the incremental NPV is equal to:

$$NPV_{B-A} = -40 + \frac{19}{1.16} + \frac{19}{1.16^2} + \frac{19}{1.16^3} = 2.67$$

Therefore, it is recommended to proceed with Project B.

Note that:

$$NPV_A = -10 + \frac{6}{1.16} + \frac{6}{1.16^2} + \frac{6}{1.16^3} = 3.48$$

$$NPV_B = -50 + \frac{25}{1.16} + \frac{25}{1.16^2} + \frac{25}{1.16^3} = 6.15$$

### 4.2.2 Projects that differ in the pattern of cash flows

Even if two projects have the same total nominal cash flow and the same initial investment, the order in which those flows arrive changes everything.

#### Example 14:

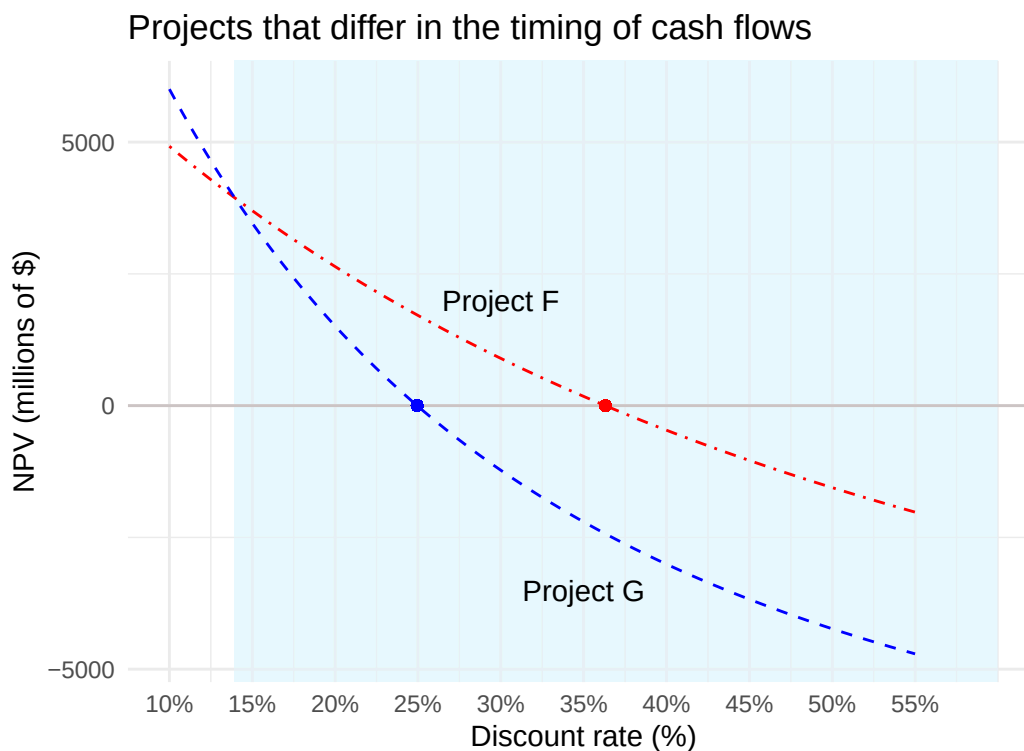
Below are the cash flows for two projects that differ in the timing of their cash flows:

| Project | 0      | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    |
|---------|--------|------|------|------|------|------|------|------|------|
| F       | -10000 | 6000 | 6000 | 6000 |      |      |      |      |      |
| G       | -10000 | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 | 3000 |

- (a) Calculate the IRR of both projects.

|     | Project F | Project G |
|-----|-----------|-----------|
| IRR | 36.3%     | 25%       |

- (b) Graph the relationship between the NPV and discount rates for projects F and G.



If the discount rate is 10%, then the NPV of Project G is greater than that of Project F. However, if the discount rate is greater than 13.9%, then the NPV of Project F is greater than that of Project G (the area shaded in light blue).

(c) Calculate the IRR of the incremental cash flows.

| Project | 0      | 1     | 2     | 3     | 4    | 5    | 6    | 7    | 8    |
|---------|--------|-------|-------|-------|------|------|------|------|------|
| F       | -10000 | 6000  | 6000  | 6000  |      |      |      |      |      |
| G       | -10000 | 3000  | 3000  | 3000  | 3000 | 3000 | 3000 | 3000 | 3000 |
| G-F     | 0      | -3000 | -3000 | -3000 | 3000 | 3000 | 3000 | 3000 | 3000 |

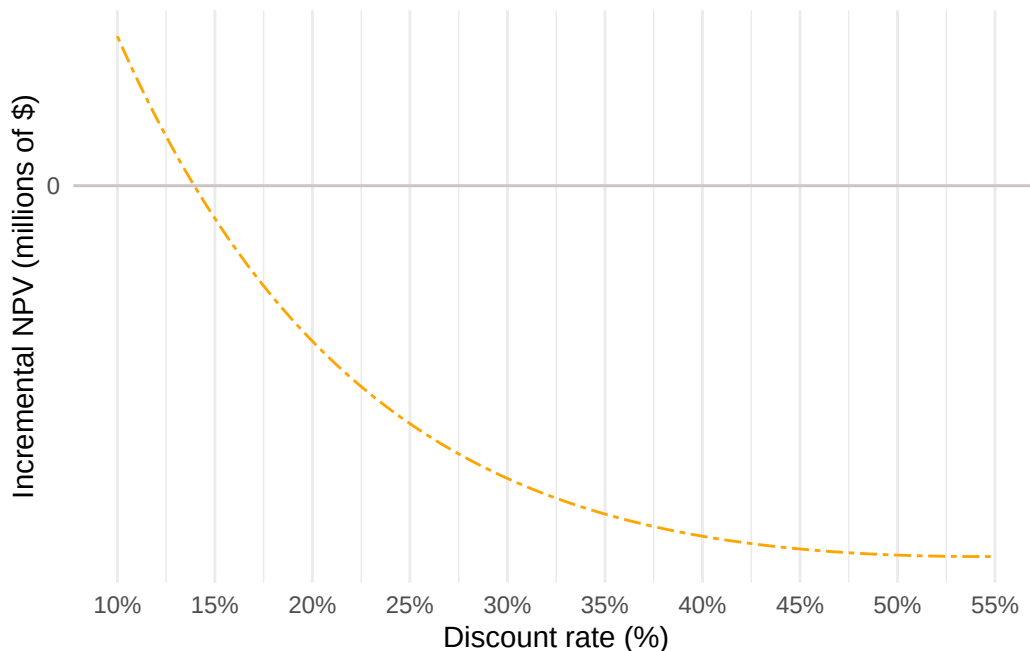
The IRR of the incremental cash flows is 13.9%. This is the point where both projects yield the exact same NPV.

*Decision:*

- At low rates ( $< 13.9\%$ ): The sheer volume of cash in Project G makes it superior.
- At high rates ( $> 13.9\%$ ): The “speed” of cash in Project F makes it the winner.

(d) If the relevant discount rate is 18%, what decision should be made?

If the discount rate is 18%, then the  $NPV < 0$ , and it is recommended not to proceed with Project G.



*It is not recommended to use the Incremental IRR as a decision criterion, as it can lead to different conclusions than the Incremental NPV in the presence of scale differences, non-conventional cash flows, or multiple internal rates of return. The correct criterion is the Incremental NPV, because it directly measures value creation and leads to the decision that maximizes the firm's value.*

## 5 Impact investment

*Impact investments are investments made with the intention to generate positive, measurable social or environmental impact alongside a financial return. (The Global Impact Investing Network (GIIN))*

*Who are the impact investors?*

The following entities own pools of capital and choose to deploy it for social or environmental impact:

- Governments
- Pension Funds and Sovereign Wealth Funds
- Angel investors
- Foundations
- Family Offices

Impact investments vary from market-rate returns (common for pension funds and foundation endowments) to concessionary/impact-first returns.

*74% of investors surveyed in the GIIN's State of the Market 2024 pursue risk-adjusted, market-rate returns, while 26% pursue below-market-rate returns.*

Impact investing offers a path to align assets with values, but the risks of early-stage ventures require precise targeting. Entrepreneurs should focus their efforts on asset managers rather than asset owners, as managers act as the direct decision-makers regarding approval time-lines, return expectations, and impact alignment. These managers range from venture capital and private equity impact funds seeking high-growth potential to nonprofit investment funds and Community Development Financial Institutions (CDFIs) that provide flexible debt for underserved markets. Furthermore, for those operating in emerging economies, Development Finance Institutions (DFIs) serve as vital intermediaries, while commercial and investment banks increasingly manage pooled vehicles for broader impact-labeled strategies.

*The Five Dimensions of Impact: Adapted from Impact Frontiers (Duke University, Impact Investment Finance)*

- **What: Impact challenge/opportunity**

What problem is the company looking to address through its impact mission?

- **Who: Target beneficiaries**

Who will experience the expected impact? (e.g. characteristics related to geography, income level, accessibility, etc.)

- **Company contribution: Contribution**

How does the company seek to solve the challenge or opportunity for the target beneficiaries? (e.g. products, services or activities)

- **How much: Expected outputs / Expected outcomes**

- What are the near term or immediate indicators that the company is delivering or its targeted contribution to the impact challenge? (*Expected output*)
- What social or environmental outcomes does the company expect as a result of its outputs? (*Expected outcome*)

- **Impact risk: Risk**

What circumstances may arise that cause the actual impact achieved NOT to align with the expected or targeted impact? How can this be mitigated?

## 6 Additional Questions

### Question 1

A company is evaluating the construction of a solar plant. Two alternatives exist:

*Project A:* Solar plant without a battery.

*Project B:* Solar plant with a battery.

Both projects use the same solar plant, and the battery cannot operate independently.

It is known that:

- The NPV of Project A is positive.
- The NPV of Project B is lower than the NPV of Project A.

What is the correct criterion to decide whether to incorporate the battery?

- A. Accept Project B only if its NPV is positive, because every project must be evaluated independently.
- B. Reject Project B automatically because its NPV is lower than that of Project A.
- C. Evaluate the incremental NPV of adding the battery and accept it only if it is positive.
- D. Compare the Internal Rates of Return (IRR) of both projects and choose the one with the higher rate.
- E. Accept both projects if their NPVs are positive, because both generate value.

### Question 2

An energy company is evaluating two investment projects with the following financial profiles:

| Project   | Description              | Initial investment | NPV         | PI  |
|-----------|--------------------------|--------------------|-------------|-----|
| Project A | 50MW Solar Plant (North) | \$20 million       | \$4 million | 1.2 |
| Project B | 40MW Wind Farm (South)   | \$10 million       | \$3 million | 1.3 |

(a) If the projects do not compete for resources, which project(s) should the company accept? Justify your answer.

The company should accept both projects. Since both Project A ( $NPV = +\$4M$ ,  $PI = 1.2$ ) and Project B ( $NPV = +\$3M$ ,  $PI = 1.3$ ) have a positive NPV, they both create value for the firm. By accepting both, the company maximizes its total value, increasing wealth by a total of \$7 million.

(b) Suppose the Board of Directors has set a strict Capital Budget of \$25 million for this year. Is your previous answer still valid?

No. With a \$25 million capital budget, the firm cannot accept both projects because their combined cost is \$30 million. The firm should choose the project with the higher NPV, which is Project A (\$4 million).

(c) If there is a third Project C that costs \$15 Million with an NPV of \$2 million, what is the optimal “portfolio” under a \$25 Million budget? Calculate the PI

| Project | Investment | NPV | PI   |
|---------|------------|-----|------|
| A       | -20        | 4   | 1.2  |
| B       | -10        | 3   | 1.3  |
| C       | -15        | 2   | 1.13 |

To find the optimal portfolio under a \$25 million budget, we must evaluate all possible combinations of Project A, B, and C that stay within that limit.

| Project | Total cost   | Total NPV   | Status             |
|---------|--------------|-------------|--------------------|
| A only  | \$20 million | \$4 million | Feasible           |
| B only  | \$10 million | \$3 million | Feasible           |
| C only  | \$15 million | \$2 million | Feasible           |
| A and B | \$30 million | \$7 million | Not feasible       |
| A and C | \$35 million | \$6 million | Not feasible       |
| B and C | \$25 million | \$5 million | Feasible - Optimal |

The optimal portfolio is Project B + Project C. Even though Project A has the highest individual NPV, the combination of B and C fits perfectly into the \$25M budget and generates a total NPV of \$5 million, which is \$1 million more than A alone.